

Shibji Shekhar Rout

MSc Advanced Computer Science | Python | LLM Systems | AI Workflow | AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with expertise in **LangChain, LangGraph, RAG, and Azure**, reducing compliance audit time by **~80%** through automated multi-modal AI pipelines. Backed by 3 years at Accenture maintaining **97% SLA** across **1M+** financial records and resolving **120+ P1 incidents**. Holds an MSc in Advanced Computer Science from the University of Leeds.

TECHNICAL SKILLS

Programming & ML: Python (Asyncio, Streaming), SQL, Scikit-Learn, XGBoost, CatBoost, PyTorch, Pandas, NumPy
AI & LLM: LangChain, LangGraph, RAG, RAG Evaluation (RAGAS, LangSmith), MCP, Graph Memory, Prompt Engineering, OpenAI API, Gemini API, Hugging Face (Transformers, Sentence-Transformers)
Deployment & APIs: FastAPI, Flask, Docker, Kubernetes, Git, GitHub Actions (CI/CD)
Cloud: Azure (Video Indexer, AI Search, ACR, App Service, Blob Storage, Application Insights)
Databases & Vector DBs: PostgreSQL, MongoDB, ChromaDB, Azure AI Search, Qdrant

EDUCATION

University of Leeds

Leeds, UK

MSc in Advanced Computer Science | Grade: Merit

Sep 2024 – Nov 2025

- **Relevant Modules:** Machine Learning, Deep Learning, Cloud Computing, Data Mining, Data Science, Algorithms

PROJECT EXPERIENCE

Brand Guardian AI: Multi-modal Video Compliance Auditing Engine [Link]

LangGraph · Azure Video Indexer · Azure AI Search · OpenAI · FastAPI · Docker · LangSmith

- Architected a multi-modal AI compliance pipeline using **LangGraph** to orchestrate video ingestion, OCR, transcript extraction, semantic RAG retrieval, and **GPT-4o** auditing against real **FTC disclosure rules** and YouTube ad specifications.
- Engineered a semantic **RAG knowledge base** by chunking and indexing real regulatory PDFs into **Azure AI Search** with OpenAI embeddings, enabling GPT-4o to retrieve and apply **FTC and YouTube compliance rules** at inference time.
- Implemented multi-modal ingestion pipeline using **Azure Video Indexer** to extract both speech-to-text transcripts and OCR on-screen text simultaneously, enabling AI auditing of spoken and visual ad content in a single pipeline run.
- Built full-stack observability using **LangSmith** for trace debugging and **Azure Application Insights** for logs and metrics.
- Detected **2 FTC compliance violations** from a YouTube ad test, returning structured JSON output with category and severity.

DocLense Finance [Link]

LangChain · ChromaDB · OpenAI · Flask · Docker · Azure · GitHub Actions

- Built a **RAG-powered** financial document Q&A system using **LangChain** and **ChromaDB**, enforcing finance-domain restriction via a structured **6-step chain-of-thought** reasoning pipeline.
- Implemented **async PDF ingestion** using **PyMuPDF** and OpenAI embeddings, with **Case ID isolation** ensuring each client document is queried independently without cross-contamination.
- Deployed Flask app containerised with **Docker**, pushed to **Azure Container Registry**, hosted on **Azure Web App**.
- Automated **two-job CI/CD pipeline** via **GitHub Actions**, triggering build and deploy on every push to main.

Student Performance Predictor [Link]

Python · Scikit-Learn · XGBoost · CatBoost · Flask · Docker · Azure · GitHub Actions

- Built an end-to-end **ML pipeline** comparing **9 regression models** using **GridSearchCV**, selecting Ridge Regression with **88.06% R²**, identifying Decision Tree overfitting at 99.97% train vs 75.72% test through rigorous model evaluation.
- Engineered modular pipeline with ingestion, transformation, and trainer components, **custom exception handling**.
- Conducted **EDA** across 1000 records, identifying lunch type and parental education as key math score predictors.

PROFESSIONAL EXPERIENCE

Accenture

Gurgaon, India

Application Development Analyst

Jul 2021 – Jul 2024

- Served as **primary client-facing technical contact** for an enterprise JDE Finance account across AUS and NZ, managing **structured discovery calls** 2–3x per week with AP and AR teams, resolving **8 tickets per month** and maintaining **97% SLA**.
- Owned **end-to-end P1 incident response** diagnosing root cause under live pressure, achieving **average 5-hour resolution**, and delivering **clear technical-to-business communication** to Accenture and client leadership across **120+ incidents**.
- Represented Accenture in **joint client-leadership meetings**, translating system constraints into actionable decisions.
- Led **technical scoping and implementation** of a JDE currency integration for India–Australia supplier payments, collaborating with **Oracle JDE team** to deliver automated **7-day exchange rate sync** across **20+ stored procedures**.
- Trusted by client AP team to **extract and deliver financial data** for annual audits during off-hours, demonstrating **deep trust**.